

Code:

```
#include <stdio.h>

#define MAXN 100005

int parent_node[MAXN];
int sz[MAXN];
int comp_count, max_sz;

int find_root(int i) {
    if (parent_node[i] == i) return i;
    return parent_node[i] = find_root(parent_node[i]);
}

int join(int i, int j) {
    int r1 = find_root(i);
    int r2 = find_root(j);
    if (r1 != r2) {
        if (sz[r1] < sz[r2]) {
            int tmp = r1; r1 = r2; r2 = tmp;
        }
        parent_node[r2] = r1;
        sz[r1] += sz[r2];
        if (sz[r1] > max_sz) max_sz = sz[r1];
        comp_count--;
        return 1;
    }
    return 0;
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    for (int i = 1; i <= n; i++) {
        parent_node[i] = i;
        sz[i] = 1;
    }
    comp_count = n;
    max_sz = 1;
```

```
for (int k = 0; k < m; k++) {  
    int a, b;  
    scanf("%d %d", &a, &b);  
    join(a, b);  
    printf("%d %d\n", comp_count, max_sz);  
}  
  
return 0;  
}
```

Input:

```
5 6  
1 2  
2 3  
3 1  
3 4  
4 5  
5 4
```

Output:

```
4 2  
3 3  
3 3  
2 4  
1 5  
1 5
```